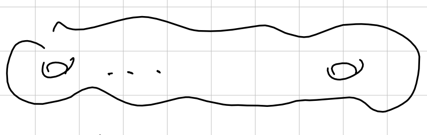
S^n, ~~$S^1\times S^1$~~, $\mathbb{RP}^n$,



($n\geq 1$)
↓
$\pi_1=0$.

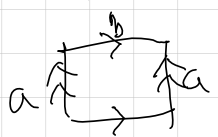
2-dim mfd

$L(p,q)$

$D^1\times S^1 / e^{2\pi i p}, e^{2\pi i q}$

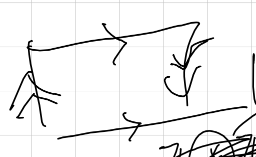
$M \vee N$. ∧ ∨

$[0,1]^2$



torus.
$\mathbb{Z}\times\mathbb{Z}$

Klein.



Klein. free product.

~~$\mathbb{Z}\times\mathbb{Z}_2$~~

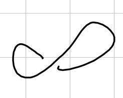
abababb..

$\langle a,b \mid aba=b\rangle$.

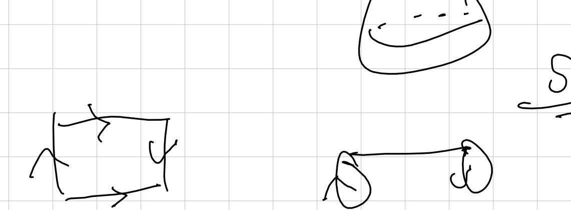
$aba^{-1}b^{-1}=e$.

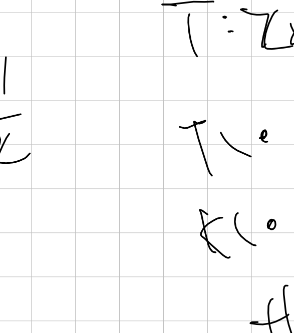
$ab=ba \to$ group $+\mathbb{Z}$.

$\langle a,b \mid ab=ba\rangle$
$\mathbb{Z}\times\mathbb{Z}$

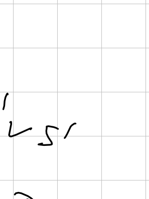


$S^1\vee S^1$
↓



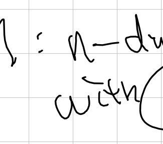


$S^1\vee S^2$



$T: \mathbb{Z}\times\mathbb{Z}$

$S^1\vee S^1$
$\mathbb{Z}*\mathbb{Z}$



$T\setminus\cdot$ $\mathbb{Z}*\mathbb{Z}$

$K\setminus\cdot$ $\mathbb{Z}*\mathbb{Z}$

theorem

Theorem

$S^1\vee S^1$
$\mathbb{Z}\times\mathbb{Z}$

S^1
↕
$\mathbb{Z}$

S^3 : 0.
↓
$\mathbb{R}^3$: 0.

M : n-dim
with ($n\geq 3$)

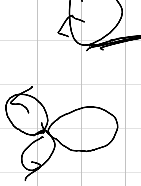
then $\pi_1(M)\simeq\pi_1(M\setminus\cdot)$.
$\simeq \pi_1(M\setminus 2pt)$
$\simeq \pi_1(M\setminus\{p_i\})$.

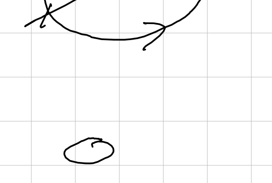
$\pi_1(S^n)=0$
$n\geq 2$.

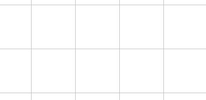
$\pi_1(\mathbb{RP}^n)=\mathbb{Z}_2$
($n\geq 2$).

tree

$S^1\vee S^1\vee\mathbb{R}$.
↓
$\mathbb{Z}*\mathbb{Z}$

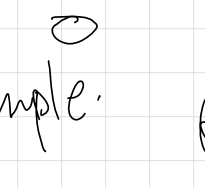


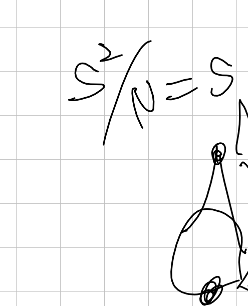




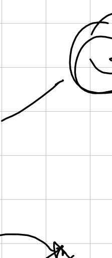
: $S^1\vee S^1\vee S^1$

$\mathbb{Z}*\mathbb{Z}*\mathbb{Z}$



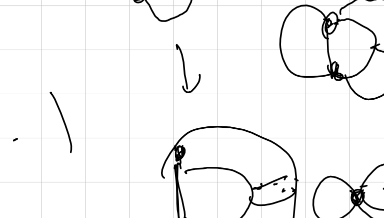


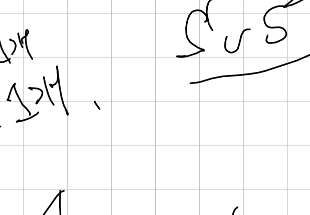
Example.

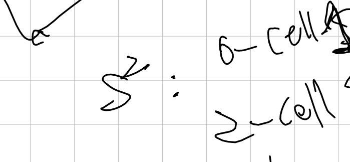


$S^2/N=S$

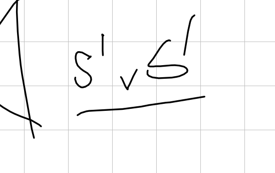
N



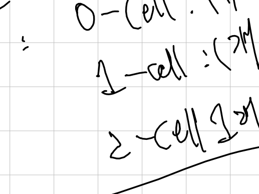


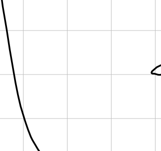


$\to \pi_1(\cdots)$



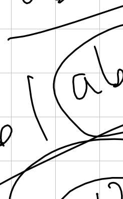
$S^1\vee S^2$,





S^1 : 0-cell 1개
1-cell 1개,

S^2 : 0-cell 1개
2-cell 1개



$S^1\vee S^1$

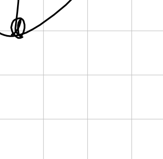
$\mathbb{RP}^2$: 0-cell : 1개
1-cell : 1개
2-cell 1개

$aba^{-1}b^{-1}=0$.

$\langle a,b \mid ab=ba\rangle$

$(\mathbb{Z}*\mathbb{Z})/\alpha$ $\mathbb{Z}\times\mathbb{Z}$

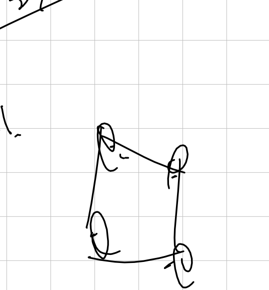
$\mathbb{RP}^2$: $\mathbb{Z}_2$ $\langle a \mid a^2\rangle$
$a^2=1$

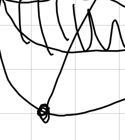


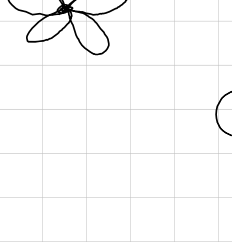
~~$\pi_1(\mathbb{RP}^3)$~~

$\mathbb{RP}^3$: ~~0-cell 1개
1-cell 1개
2-cell 1개
3-cell 1개~~

$\pi_1(\mathbb{RP}^3)=\pi_1(\mathbb{RP}^2)=\mathbb{Z}_2$.

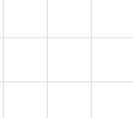


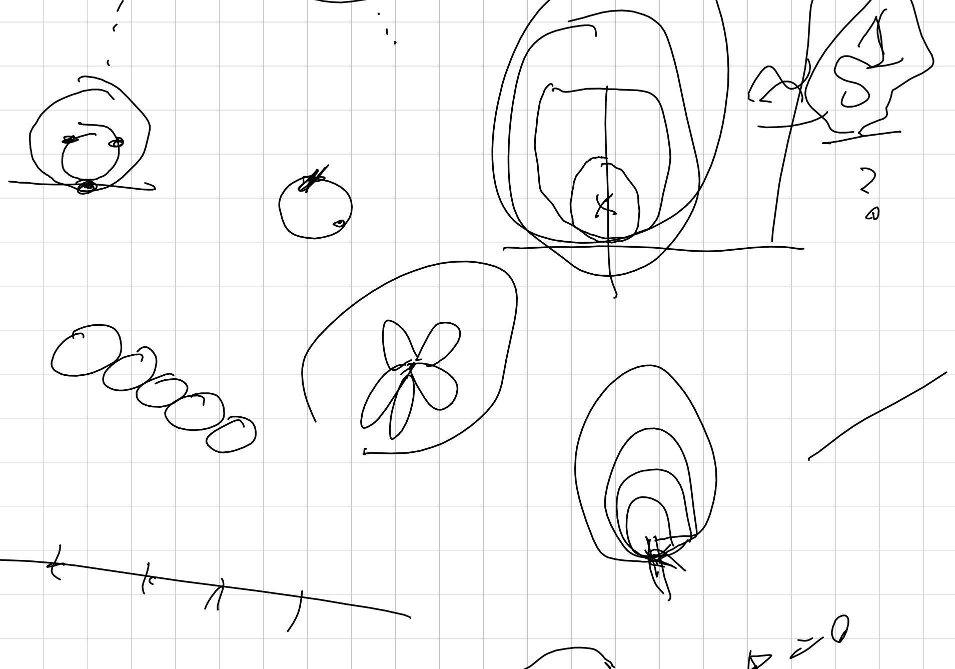




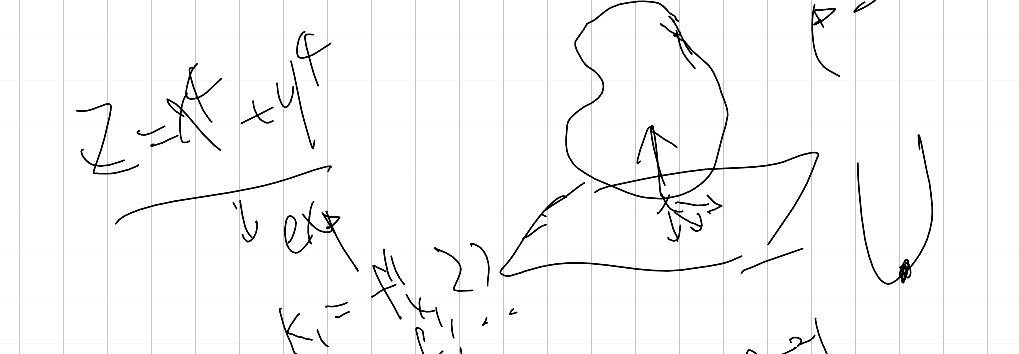
← $x^2+y^2+z^2=1$
∪ $x^2+y^2\leq 1$
∪ $(0,0,z)$ $-1\leq z\leq 1$.

$(\frac{1}{2},\frac{1}{2},\frac{1}{2})$



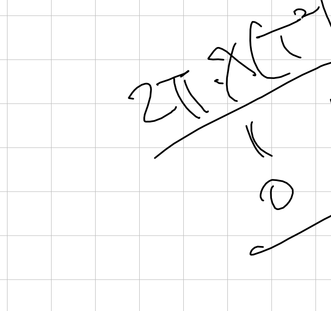


$\mathbb{R}/\mathbb{Z}$



$r=0$

$z=x^4+y^4$
↓ exp



$K_1=f(4,2)$..
$K_2=f_4$..

$\frac{1}{r^n+z}$.

$\int\frac{R\,dz}{y}$ $\int\frac{1}{z}\,dz$
$2\pi i$

$2\pi i\cdot x(z^{-1})$
0